

Recommendations to Improve Usability and The HCI Challenge of Generative AI: Designing Credible and Effective Simulations for Deliberate Practice in Special Education

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Abstract. Initial Teacher Education (ITE) faces the chronic challenge of bridging the gap between theoretical knowledge and situated classroom practice. This paper analyzes the impact of Generative Artificial Intelligence (GenAI) on the design of dynamic simulations for Special and Inclusive Education. Unlike traditional systems based on rigid scripts, the proposed ecosystem utilizes Large Language Models (LLMs) to model synthetic agents—Teo and Jojo—and provide automated formative assessment. Through a mixed-methods pilot study (N=17), a pedagogical cycle of "planning, implementing, evaluating, reflecting, and adjusting" was assessed. The quantitative results reveal an "Excellent" usability score (SUS = 80.0) and a significant increase in the participants' pedagogical confidence, rising from 3.53 to 4.29. Qualitative analysis highlights that the AI's ability to simulate not only a diagnosis but also the student's identity and vulnerability generated a credible "social presence" and a deep empathetic response. It is concluded that GenAI redefines the interface of teaching practice, establishing a "safe space for freedom to fail" that fosters metacognition and the construction of professional identity. The system enables pre-service teachers to act as conductors of human development, facilitating pedagogical adjustment and decision-making in highly complex inclusive environments.

Keywords: Generative AI, Initial Teacher Education, Inclusive Education, Pedagogical Simulation, HCI, Deliberate Practice.

1 Introduction

Initial Teacher Education (ITE) faces a chronic structural challenge: the dissonance between the theoretical pedagogical knowledge acquired at university and the situated, complex, and unpredictable reality of the school classroom [1]. This "theory-practice gap" is particularly critical in Special and Inclusive Education, where pre-service teachers often report feeling inadequately prepared to implement high-leverage practices (HLPs) required to serve students with diverse needs [2].

Historically, simulations have emerged as a vehicle for providing "deliberate practice" in safe environments. However, previous generations of these tools, based on decision trees and predefined scripts, have been criticized for their rigidity and lack of ecological validity [3]. The emergence of Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) promises to transform this landscape, allowing for the design of dynamic pedagogical agents capable of sustaining fluid and unscripted interactions [4].

This article analyzes the impact of the transition from script-based systems to dynamic interaction interfaces driven by GenAI. Specifically, we examine how autonomous agents can simulate complex student profiles for teacher training, addressing design challenges in Human-Computer Interaction (HCI) such as anthropomorphism and credibility, as well as the psychometric validity of LLMs for automated formative assessment [5].

2 Related Work

Recent literature places this study at the intersection of advanced educational technology, learning psychology, and HCI. Below, we review key advances in generative simulations and automated assessment.

2.1 Evolution of Simulations: From Scripts to Dynamic Agents

The transition toward generative environments represents a paradigm shift. Traditional simulations, limited by pre-programmed responses, often failed to replicate the conversational complexity of the classroom, leading to mechanical learning [3]. In contrast, recent research demonstrates that agents powered by LLMs can exhibit improvised and adaptive behaviors, facilitating deliberate practice cycles where teachers refine their skills through repetition with variation [4]. In the context of inclusive education, these systems allow for the personalization of scenarios to train in managing disruptive behaviors and curricular adaptation—areas where traditional field practice is often insufficient [6].

2.2 HCI Challenges: Credibility and Anthropomorphism

The pedagogical effectiveness of a virtual agent depends on its ability to evoke a social response. Social presence theory suggests that anthropomorphism can facilitate immersion; however, recent studies warn of its risks. Reinecke et al. [7] argue that anthropomorphism in LLMs is a "double-edged sword": it can increase engagement but also lead to over-reliance on the system's accuracy or "technical hallucinations". To

maintain credibility, it is crucial that agent design aligns appearance and voice with simulated cognitive abilities, avoiding the dissonance that breaks immersion [8].

2.3 LLM Validity for Automated Formative Assessment

An essential component of deliberate practice is immediate feedback. The authors of study [5] evaluated the reliability of LLMs (including GPT-4 and open-source models) for automated scoring in formative assessments. Their findings indicate that premium models, used with appropriate prompting techniques, reach levels of correlation with human evaluators comparable to traditional methods, while also offering detailed explanations [5]. This suggests that GenAI can effectively scale formative feedback, although human supervision is required to mitigate potential biases in high-stakes assessments [9].

3 Proposed Solution: GenAI-Based Pedagogical Simulation System

The proposed solution consists of a web-based interactive simulation platform designed to bridge the gap between theoretical training and real pedagogical practice in Special Education. Unlike traditional systems based on decision trees or linear scripts, this proposal leverages Generative Artificial Intelligence (GenAI) to model complex and fluid human behaviors. This approach enables a "safe space for freedom to fail," where future teachers can develop critical skills such as evaluating, designing, implementing, managing, deciding, and self-regulating without negative consequences for real students, thereby fostering "deliberate practice" [10].

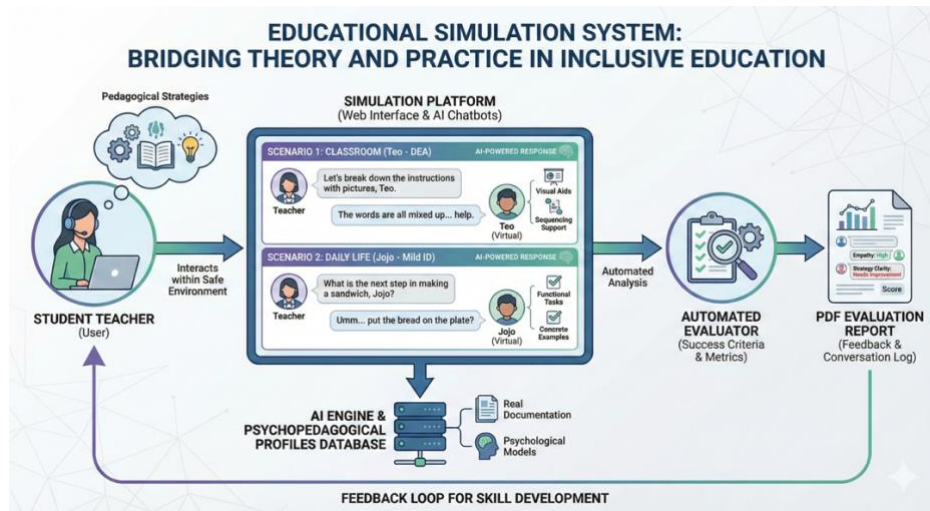


Fig. 1. Conceptual Architecture of the Pedagogical Simulation System.

3.1 General System Architecture

As shown in Fig. 1, the system operates as an integrated ecosystem that connects pre-service teachers with virtual agents through a formative feedback loop. The architecture is organized into three functional layers:

- **Interaction Layer:** Users interact through natural language with chatbots that assume specific roles representing students with Special Educational Needs (SEN). This layer aims to replicate the sensory and cognitive load of a real mediation.
- **User Intelligence Layer (User Modeling):** A GenAI engine processes the conversation based on a database of profiles constructed from real psychopedagogical and psychological documentation. Here, the AI does not act as a content generator but as a model of human behavior.
- **Formative Assessment Layer:** An analytical module objectively examines the interaction, contrasting it against inclusive best-practice criteria to generate immediate feedback reports, acting as a "reflective mirror" for the teacher.

3.2 Conceptual Foundation: The Multidimensional Model

The core of the system is not based on static diagnostic descriptions, but on the Multidimensional Model of Disability. This approach allows the simulated student to be understood as a dynamic interaction among multiple dimensions, avoiding reductionist perspectives:

- **Cognitive and academic skills:** Defines how the agent processes information and responds to specific curricular content.
- **Emotional regulation and adaptive behavior:** Models reactions to frustration, error, or social pressure, enabling observation of the teacher's emotional validation strategies.
- **Family and social context:** Incorporates environmental factors that influence learning disposition and participation.

3.3 Design of Simulated Student Agents

The use of virtual platforms powered by generative AI for teacher training enables immersive and highly realistic training, facilitating the special education teacher's role as a conductor of development through problem-solving in simulated and dialogic environments. Two differentiated profiles were developed based on real psychopedagogical reports prepared by experts:

- **A. "Teo" Profile (Specific Learning Difficulties - SLD):** Simulates a 9-year-old child (4th grade) with literacy difficulties. Its design is structured across four levels: Condition, Characteristics (skills and preferences), Behavior (responses to pedagogical strategies), and Presentation (age-appropriate language). Teo responds positively to visual supports and continuous validation but exhibits avoidance behaviors when facing abstract language-based tasks.
- **B. "Jojo" Profile (Mild Intellectual Disability - MID):** Represents a 15-year-old student (9th grade). The modeling focuses on concrete thinking and the Transition

to Adult Life (TAL) approach. This agent enables practice in reducing curricular abstraction and linking learning objectives to functional, everyday life situations.

3.4 The Automated Pedagogical Evaluator

This module plays a fundamental and complementary role to the pedagogical chatbot, acting as a non-punitive expert observer that allows for closing the reflective learning cycle. While the agent simulates the behavior of a student with SEN, the evaluator analyzes the teacher's mediation to avoid a merely instrumental interaction and foster metacognition regarding the strategies deployed.

The evaluation is based on the principles of situated teaching and Evidence-Centered Design (ECD), operating under the principle of strict observational evidence: the system objectively analyzes only the behaviors and messages explicitly present in the chat log, without inferring intentions not textually documented. This approach ensures that the feedback provided is effective, based on the premise that feedback is one of the factors with the greatest impact on learning and professional performance improvement. The criteria were defined based on Special Education methodological guidelines, functional and ecological scaffolding principles, and emotional validation strategies aligned with the Multidimensional Model of Disability.

The 11 criteria that make up the system's evaluation matrix are detailed below, ensuring the simulation aligns with inclusive practices and real Special Education contexts:

Table 1. Automated Pedagogical Evaluator Criteria Matrix.

Criterion	Description and Observable Indicators
1. Use of Functional/Ecological Scaffolding	Teacher's ability to link the session topic to everyday life situations and real contexts (e.g., shopping, transport).
2. Clear Step Sequencing	Decomposition of complex tasks into simple, sequential instructions to reduce the student's cognitive load.
3. Adaptation of Texts and Prompts	Simplification of language and elimination of abstractions or ambiguities to ensure information accessibility.
4. Use of Memory for the Concrete	Use of elements from the student's memory or tangible examples to reinforce the bond and attention.
5. Prevention of Mockery and Fear	Implementation of strategies to reduce anxiety and prevent fear of error or mockery during learning.
6. Validation of Vulnerability	Ability to recognize and validate the student's emotional state or frustration before proceeding with a pedagogical correction.
7. Promotion of Social Autonomy	Encouragement of decision-making and active participation of the student in simulated social contexts.
8. Ecological Curricular Linkage	Alignment of pedagogical objectives with the student's functional needs and natural environments.
9. Early Vocational Inquiry	Incorporation of the student's interests, tastes, and preferences to motivate learning toward future goals.

- | | |
|-----|---|
| 10. | Reinforcement of Promotion of skills that facilitate the student's inclusion and Community Autonomy independent performance in their community. |
| 11. | Promotion of Curricular Implementation of the necessary supports and adjustments to Inclusion ensure equitable access to the curriculum. |
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The evaluator is automatically activated upon completion of the interaction, generating structured and comprehensible feedback in a PDF report. This report includes the full conversation log, a criteria compliance table, a qualitative performance analysis, and a global conclusion with suggestions for improvement. This design transforms the simulation into a comprehensive learning experience, allowing the teacher to practice inclusive strategies and receive immediate feedback based on their own practice.

3.5 Simulation Outcomes: The Formative Report

The cycle concludes with the generation of an Evaluation Report in PDF format. This document functions as a metacognitive tool that triggers the reflection and adjustment process, allowing students to analyze their own performance through three sections: a complete interaction log for tone analysis, a compliance analysis that justifies success or failure based on textual evidence, and a narrative synthesis with practical suggestions for future simulation sessions.

4 System Implementation and Architecture

This section details the technological ecosystem and user interface (UI) design that support pedagogical interaction. The solution was built using a decoupled microservices architecture to ensure conversational fluency and accuracy in evaluative analysis.

4.1 Software Architecture and Technology Stack

The platform adopts a modern stack optimized for the integration of large language models (LLMs) and real-time interaction management.

Backend and Processing: FastAPI (Python) was implemented as the main framework due to its high performance and asynchronous handling capabilities. This choice is critical for reducing AI response latency while maintaining the conversational coherence required in educational simulations, ensuring that the interaction remains efficient and minimizes the user's cognitive load.

Data Management: Data persistence is handled through PostgreSQL and the SQLAlchemy ORM. This infrastructure enables the complete storage of conversation histories, ensuring the traceability required by the evaluation module.

Artificial Intelligence Engine: The system core uses the Google Gemini API (model gemini-2.0-flash). Independent prompts were designed to operate under three distinct roles: Teo simulation, Jojo simulation, and the automated pedagogical evaluator. This structure allows the LLM to generate responsive and pedagogically aligned behaviors for deliberate practice, reinforcing the separation of functions between simulation and assessment.

Infrastructure and Frontend: Deployment is managed through Docker to ensure consistency across execution environments. The user interface was developed using React and Tailwind CSS, prioritizing simplicity and minimizing user cognitive load to keep the focus strictly on the pedagogical challenge.

4.2 User Interface and User Experience Design (UI/UX)

The development of the technological ecosystem was based on agile methodologies and user-centered design (UCD) processes, ensuring that the information architecture (IA) was flexible enough to adapt to learning dynamics. Specifically, the design process integrated framework guidelines that allow for an agile integration of information architecture into the development flow, prioritizing end-user satisfaction and reducing cycle times by obtaining usable and validated software increments. Under this approach, the interface design follows a logical flow oriented toward reflective learning, from institutional access to final feedback (See Fig. 2).

A. Welcome Screen and Institutional Access

The landing page presents an intuitive design aimed at introducing users to the pedagogical approach of the tool. The authentication system functions as an institutional access control mechanism exclusive to Universidad San Sebastián.

Strict Validation: Access is restricted to email domains correo.uss.cl or docente.uss.cl. This restriction responds to ethical criteria to ensure the tool is used within a controlled university training context.

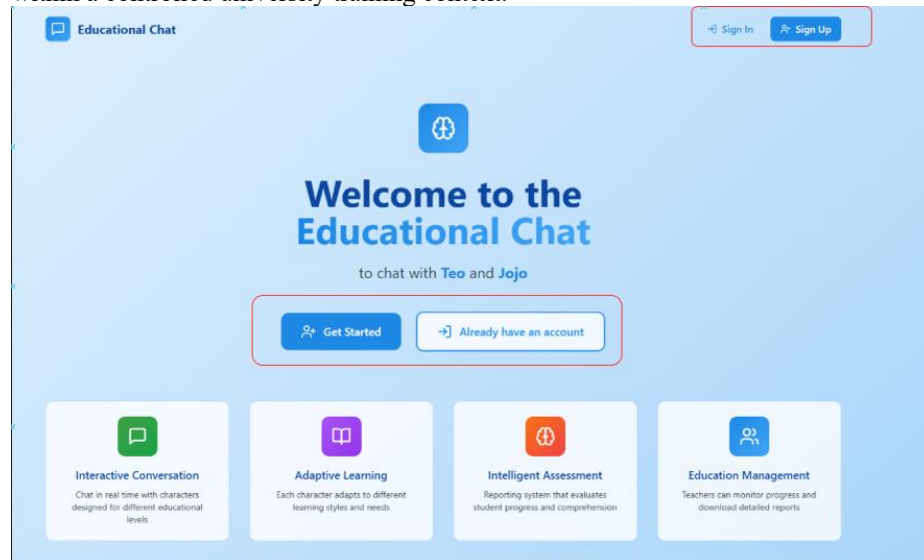


Fig. 2. Welcome Interface and Institutional Access Control.

B. Scenario Selection and Pedagogical Scaffolding

After authentication, users access a selection screen to choose the practice profile: Teo or Jojo (See Fig. 3).

Contextualization: Each profile includes age, educational level, and a psychopedagogical description that guides the teacher regarding the challenges to be addressed.

Integrated Scaffolding: At the bottom of the screen, the section “*Tips for a better conversation*” is included. These tips function as preliminary pedagogical scaffolding, suggesting the use of visual supports and pedagogical patience before initiating the dialogue.

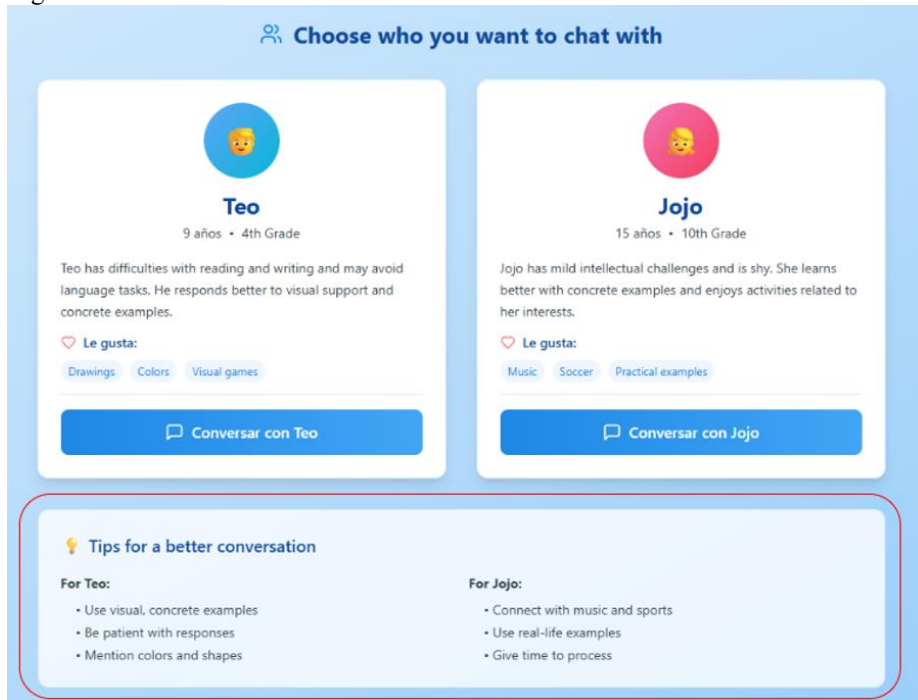


Fig. 3. Scenario and Simulated Profile Selection Screen.

C. Interaction Interface and Saving Functionality

The chat environment enables real-time communication with the AI-based assistant. The interface is intentionally minimalistic to focus user attention on conversational fluency (See Fig. 4).

“Finish and Save” Button: This functionality is critical; upon completing the session, the dialogue is automatically stored to feed the evaluation and tracking modules.

D. Evaluation and Metacognition Module

The system concludes with an Evaluation Report view, designed to close the learning cycle through reflection on teaching practice.

Interaction Log: Presents a complete participant/message transcript, allowing users to review tone and the types of questions posed (See Fig. 5).

Evaluation Table: Structures the analysis according to pedagogical criteria (e.g., step sequencing, text adaptation), indicating compliance, qualitative analysis, and justification based on concrete evidence from the conversation (See Fig. 6).

Performance Range and Feedback: The system assigns a score and a performance range (Acceptable, Competent, Successful). It concludes with a narrative summary identifying strengths and actionable areas for improvement in future sessions.

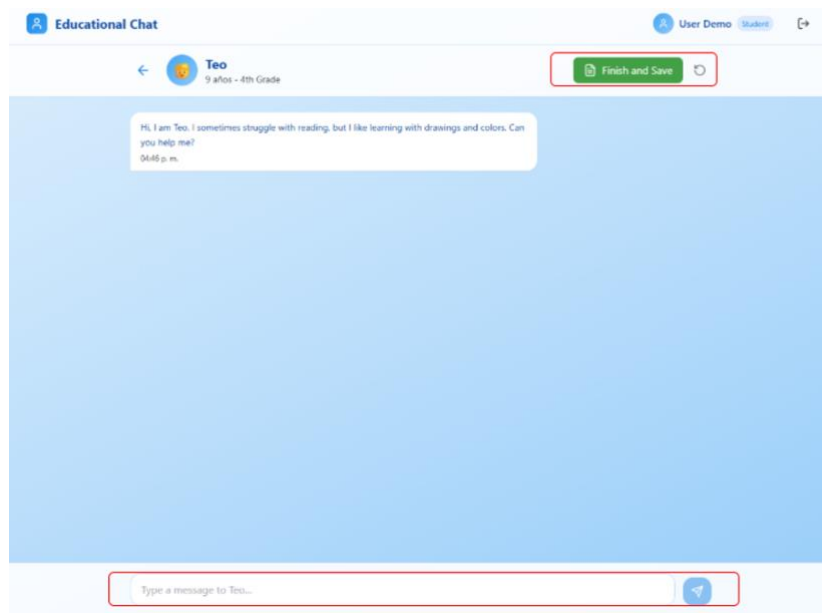


Fig. 4. Real-Time Interaction Environment (Educational Chat).

Preview: Evaluation Report Download PDF

Teo
9 years - 4th Grade
Date: 25-01-2026

1. Interaction

Participant	Message
Teacher	Hi Teo! Welcome to a new session. I'm special education teacher María de los Angeles. How are you feeling today?
Teo	Hi, teacher María. I'm okay.
Teacher	That's great, Teo! How is your family and your dog Rufino?
Teo	They're good. Rufino is sleeping.
Teacher	He's probably tired. Did you go to the park today?
Teo	No, not today. My dad was busy. Rufino just sleeps.
Teacher	I have an idea for what we could do today that's fun and based on the adventures you have with your grandma. Want to know what it is?
Teo	Oh! Yes, tell me.
Teacher	Let's draw, but besides that, I want to suggest something. How about writing a very short story about what you and your grandmother did in the park?
Teo	Draw? Yes! But... a story? I don't know. My handwriting is messy.
Teacher	Don't worry, everyone struggles with writing at some point, even me. I'm here to help. Anyway, if you'd like, we could invite one of your friends to help us with this activity. What do you think of that idea?

Fig. 5. Formative Assessment and Performance Reporting Module.

Teo Okay! Um... My grandma and I went to the park and we saw a big dog, like Rufino. And we drew ducks.

2. Evaluation Table

Criterion	Description	Compliance	Analysis	Justification
1. Use of Functional/Ecological Scaffolding	The teacher attempts to link the session topic to a real-life situation (e.g., shopping, transport, etc.).	YES	The teacher proposed an activity directly based on the student's personal experiences with his grandmother in the park, linking learning to his real context.	Evident in messages: 'I have an idea for what we could do today that's fun and based on the adventures you have with your grandma.' and 'How about writing a very short story about what you and your grandmother did in the park?'
2. Clear Step Sequencing	The teacher breaks the activity into simple visual steps and avoids complex instructions.	YES	The teacher provided clear, sequential instructions for how the activity would proceed with the student and his friend.	Evident in message: 'Perfect. So, tell Pedro what you did with your grandmother in the park. Then, with my help, we'll write the short story, and Pedro will be in charge of reading it.'
3. Adaptation of Texts and Prompts	The teacher simplifies language and avoids abstract questions.	YES	The teacher consistently used simple, concrete language and questions appropriate for the student's age and grade level, avoiding any abstract concepts.	Evident in messages like 'How are you feeling today?', 'Did you go to the park today?', and 'How about writing a very short story?'

Fig. 6. Evaluation table Module.

5 Methods

A mixed-methods (convergent) descriptive pilot study was conducted, designed to evaluate the usability, interaction credibility, and formative effectiveness of a GenAI-based simulation ecosystem. The study integrated quantitative metrics of usability and pedagogical confidence with a qualitative analysis of the user experience.

5.1 Participants and Selection Criteria

The sample consisted of N=17 students from the Special Education program at San Sebastián University (Chile). Participation was voluntary and based on the following inclusion criteria: (a) being regularly enrolled in the Special Education program, (b) having an active institutional email account, and (c) currently participating in intermediate or final internships.

- **Demographic Profile:** The sample was entirely female, with an average age of $M=23.35$ years ($SD=3.23$, range 20-33). In terms of academic progression, 35.3% were in their third year, 29.4% in their fourth year, and 23.5% in their fifth year, with a minority representation of first and second-year students.
- **Digital Competence and Previous Experience:** Participants reported a perceived digital competence level of $M=4.12$ ($SD=0.6$) on a scale of 1 to 5. Notably, 70.5% of the sample ($n=12$) declared having previous experience with Generative Artificial Intelligence tools prior to the intervention.
- **Sample Justification:** Although the sample size is limited, it is statistically appropriate for formative usability evaluations. Specialized HCI literature indicates that between 10 and 12 participants are sufficient to detect 80% of usability problems and obtain precise measurements on the SUS scale.

5.2 Materials and Simulation Environment

The experimentation was carried out using a web platform specifically designed for this purpose (for more details, see sections 4 and 5). The materials included:

1. **Simulated Agents (User Models):** Two pedagogical profiles were configured based on real psychopedagogical documentation: Teo (9 years old, 4th grade, diagnosed with SLD in literacy) and Jojo (15 years old, 9th grade, diagnosed with Mild Intellectual Disability).
2. **Student Files:** Participants received printed or digital documentation (via QR code) with pedagogical background, interests, and support needs for each agent, simulating access to a real "student folder".
3. **Technical Infrastructure:** The environment was run on the students' personal devices under a controlled university network, accessing via institutional authentication to ensure data traceability.

5.3 Data Collection Instruments

A set of standardized and ad-hoc instruments was used to capture the multidimensionality of the experience:

- **System Usability Scale (SUS):** A 10-item questionnaire (5-point Likert scale) to measure the perceived usability of the system, internationally validated for its high reliability.

- **Psychological Perception Scales:** Three constructs were measured using 1 to 5 scales: (1) Initial level of anxiety toward technology, (2) Pedagogical confidence before the simulation, and (3) Pedagogical confidence after the intervention.
- **Qualitative Reflection Questionnaire:** A 12-question open-ended survey aimed at evaluating agent credibility (social presence), response authenticity, the usefulness of the feedback report, and preference for interaction modalities (text vs. voice).

5.4 Experimental Procedure

The study was structured under an 85-minute protocol, designed to activate a pedagogical cycle of reflective iteration based on five key stages: plan, implement, evaluate, reflect, and adjust. This design aims for the pre-service teacher not only to interact with the technology but also to develop deliberate practice and a conscious professional identity. The procedure was divided into the following timed phases:

1. **Phase 0: Onboarding and Technological Safety (5 min):** Registration and free exploration of the platform. This stage aims to reduce novelty bias and mitigate the initial anxiety detected in the participants' profile, establishing a technical trust environment.
2. **Phase 1: Pedagogical Planning (Plan) (20 min):** Participants analyzed the clinical and pedagogical file of the assigned agent (Teo or Jojo). They drafted an "Intervention Plan v1," defining learning objectives and a structure for the beginning, development, and closure, thus avoiding improvisation in mediation.
3. **Phase 2: First Implementation and Evaluation Cycle (35 min):**
 - **Implementation (15 min):** Execution of plan v1 through dialogic interaction with the chatbot.
 - **Evaluation and Reflection (5 min):** Analysis of the feedback report generated by the AI. At this point, the platform acts as a "reflective mirror," allowing the student to identify gaps in their pedagogical mediation.
 - **Adjustment (15 min):** Redesign of the intervention plan, generating the "Plan v2" version strictly based on the evidence of previous performance and the system's suggestions.
4. **Phase 3: Second Cycle and Consolidation (25 min):**
 - **Re-implementation (15 min):** Second interaction with the agent to apply the planned adjustments and observe the evolution of the virtual student's response.
 - **Exit Evaluation (10 min):** Closure of the session and response to the measurement instruments (SUS and qualitative questionnaire) to capture the final perception of self-efficacy and the user experience.

6 Results

A mixed-methods analysis approach was adopted to evaluate the effectiveness of the system. Quantitative results for usability and pedagogical confidence are presented

first, followed by a thematic qualitative analysis that delves into the students' perceptions of the tool and their interaction with the AI agents.

6.1 Perceived Usability Analysis (SUS)

The usability of the simulation ecosystem was evaluated using the System Usability Scale (SUS), applied to the total sample of participants (N=17) after completing the two pedagogical intervention cycles.

Quantitative Results and Benchmarking. The global average score obtained was $M = 80.0$ ($SD = 12.4$), with a median of 82.5 and a range spanning from 50.0 to 95.0. According to the reference frameworks established by [19], [20], a score of 80.0 places the platform in the 90th percentile, granting it an "Excellent" acceptability grade and an "A-" letter grade. These results indicate a high readiness among users to adopt the system in real professional training contexts.

A detailed analysis of the 10 SUS items revealed critical strengths in the "Learnability" and "Consistency" dimensions:

- **Frequency of Use and Learning:** The items with the highest positive scores were Q1 ("I think that I would like to use this system frequently", $M=4.53/5$) and Q7 ("I would imagine that most people would learn to use this system very quickly", $M=4.53/5$).
- **Complexity and Support:** Participants perceived low technical complexity, reflected in low averages for Q8 ("I found the system very cumbersome to use", $M=1.59/5$) and Q4 ("I think that I would need the support of a technical person to be able to use this system", $M=1.94/5$), suggesting a minimal learning curve for pedagogy students.
- **Integration:** The perception that functions were well integrated (Q5) reached a mean of 4.41/5, validating the technical architecture that connects the chat with the evaluator module.

Qualitative Perception of the Interface and Chat Format. The analysis of open-ended responses in the qualitative questionnaire supports the high SUS scores, identifying the written chat format as a facilitator of cognitive clarity. Students highlighted three main benefits of the interface:

1. **Reduction of Improvisation:** The format allowed users to maintain focus on the previous pedagogical plan. One participant noted: *"The written chat format helped me to order my ideas and use clear language"* (P2), while another emphasized that it *"helped to organize ideas before writing them... and correct things that might cause confusion with audio"* (P3).
2. **Ease of Use and Friendliness:** The platform was described as *"quite friendly and comfortable"* (P4) and *"clear and very easy to use"* (P5). It was especially valued that the interaction felt *"so natural that it felt like talking to a child"* (P5).
3. **Opportunities for Improvement in Stability and Multimodality:** Despite the positive evaluation, design needs were identified for future iterations. Users suggested incorporating visual supports (images) to favor scaffolding (P1, P3) and

the possibility of voice interaction to increase situational realism (P5, P7). Additionally, the need to improve system stability against specific errors that interrupted interaction was mentioned (P9).

6.2 Impact on Pedagogical Confidence (Quantitative Results)

The effect of the simulation on the students' perception of self-efficacy was evaluated by comparing their confidence levels before and after the intervention. The results show a significant positive transition in the participants' professional security:

- **Pre-Intervention Analysis:** Before interacting with the ecosystem, students reported a mean pedagogical confidence of $M=3.53$ ($SD=0.72$) on a 1 to 5 scale. This baseline level correlates with a moderate initial anxiety level ($M=2.94$), reflecting the typical uncertainty of pre-service teachers when facing complex SEN cases without prior training.
- **Post-Intervention Analysis:** After completing the two simulation cycles and receiving feedback from the automated evaluator, pedagogical confidence rose to a mean of $M=4.29$ ($SD=0.59$).
- **Magnitude of Change:** A net increase of +0.76 points was observed on the scale. Notably, 82.3% of participants ($n=14$) reported the maximum confidence level (4 or 5) in the exit phase, compared to only 52.9% in the initial phase. This finding suggests that the iterative structure of the task (Planning / Execution / Feedback) functions as an effective scaffolding mechanism for pedagogical decision-making.

6.3 Agent Credibility and Pedagogical Empathy (Qualitative Analysis)

The qualitative analysis allowed for the exploration of the "Social Presence" dimension and the pedagogical validity of the user models. This section delves into the agents' fidelity in generating a simulation linked to school reality.

Model Coherence and Situational Realism (Fidelity). One of the most relevant findings was the perception of the agents' authenticity, although the analysis identified critical nuances between emotional and academic dimensions:

- **Behavioral and Emotional Consistency:** Participants positively evaluated the emotional and behavioral coherence of both agents relative to the profiles described in their background. It was noted that the agents respond appropriately to their diagnosis and age, allowing for a credible and fluid interaction.
- **Nuances in Academic Fidelity:** Despite behavioral consistency, discrepancies were identified in Teo's academic performance. A group of participants perceived that this agent showed higher competencies than expected for a profile with Specific Learning Difficulties (SLD). This finding reveals that, while the behavior is credible, the level of academic performance requires fine-tuning in the prompt design to achieve more precise diagnostic calibration.
- **Social Presence:** The interaction was described as natural, similar to talking to a real child, which avoided the "uncanny valley". This fluidity allowed users to immerse themselves in their professional role of facilitating inclusion processes.

Affective Responses and Professional Empathy. The simulation managed to evoke a complex emotional curve reflecting real practice experiences:

- **Emotional Transition:** Students reported moving from nervousness and the "*fear of not doing it right*" toward a state of tranquility once the interaction was established. This pattern is similar to that experienced in real educational centers.
- **Connection and Pedagogical Empathy:** The system generated deep empathetic responses. The AI's ability to simulate the student's self-concept was particularly impactful; users mentioned feeling "*anguish and sorrow*" when the agent expressed frustration by comparing himself to his siblings.
- **Realism in Frustration:** The virtual student's moments of fatigue and frustration were perceived as real reactions, allowing for the practice of emotional validation in tense situations.

The Interaction Cycle as a Didactic Trigger. The analysis confirmed that the ecosystem acts as a catalyst for the process of planning, implementing, evaluating, reflecting, and adjusting:

- **Formative Assessment:** The evaluation report was perceived as a strategic map for identifying areas for improvement. However, it was acknowledged that occasional technical glitches could generate "false positives or negatives" in the assessment, which were identified by the participants through their professional judgment.
- **Higher-Order Skill Development:** The iterative cycle promoted self-analysis and pedagogical decision-making skills. Students valued the ability to contrast their internal perception with the system's objective evaluation to perform better interventions.

6.4 Modality Preferences and Improvement Projections

The analysis explored how system configuration influences the development of teaching skills and which elements would enhance its effectiveness in inclusive training.

Text vs. Voice Interaction: The Reflection-Spontaneity Binomial. A distinct pedagogical function was identified for each modality:

- **Written Chat as Cognitive Scaffolding:** The written format was positively valued for favoring "focus, planning, and concrete pedagogical decision-making". It allowed users to "reflect, organize ideas, and maintain focus on the intervention plan," avoiding improvisation and forcing a more precise and structured use of language.
- **Voice as a Factor for Realism and Adjustment:** There is a consensus (n=15) that voice would enrich the experience by favoring a "more spontaneous response closer to classroom reality". The possibility of "hearing how a student with SEN expresses themselves" and detecting emotional nuances in voice tone would allow practicing pedagogical adjustment in real-time in a more organic way.

Temporality, Functionality, and Visual Scaffolding. Participants proposed adjustments to deepen the tool's impact:

- **Time Management:** The 15-minute interaction was described as "interesting and challenging," being suitable for training specific objectives. However, it was suggested to complement this with an open-ended modality to delve into complex situations or extend pedagogical reflection.
- **Multimodal Support:** The need to include visual supports was reiterated. The ability to "add documents or images" would allow for more inclusive interventions suited to each student, facilitating the necessary scaffolding in SLD and MID profiles.
- **Progress Traceability:** To strengthen autonomy, a "conversation history" was suggested to observe longitudinal progress, facilitating sustained self-reflection on the development of their own teaching competencies over time.

7 Discussion

The findings of this pilot study demonstrate that the integration of GenAI-powered Synthetic Agents, when structured under a multidimensional model, is not only technically feasible but profoundly alters the dynamics of initial teacher education. Unlike traditional simulations, this ecosystem does not merely reproduce behaviors; it triggers a complete didactic process: planning, implementing, evaluating, reflecting, and adjusting. The following sections discuss the implications of these results regarding the challenges of credibility, metacognition, and interface design within the scope of Inclusive Education.

7.1 Social Presence and the Simulation of Student Identity

The success of an educational simulation in HCI depends on the user's ability to suspend disbelief and treat the agent as a legitimate social entity [21].

- **Beyond Logical Consistency:** The four-level prompt design succeeded in mitigating the risk of technical inconsistencies. However, qualitative analysis identified a critical area for improvement: while behavior and emotion were credible, the agent "Teo" exhibited academic competencies superior to those expected for an SLD diagnosis. This finding suggests that while the agent's humanization and a clear projection of self-concept were achieved, diagnostic fidelity in academic terms requires a fine-tuning process of the prompts to avoid discrepancies with the clinical profile.
- **Emergence of Professional Empathy:** This simulated vulnerability, despite the academic nuances, acted as the engine for pedagogical empathy. The results indicate that AI can simulate the affective dimension of learning, generating an authentic emotional connection that places the teacher in their role as a facilitator of human development.

7.2 Metacognition in a "Safe Space for Freedom to Fail"

Using AI as a "reflective mirror" allowed for overcoming the limitations of static simulations, providing what we define as a safe space for freedom to fail.

- **The Emotional Learning Curve:** Qualitative analysis revealed a critical transition from initial anxiety and the "fear of not doing it right" toward a state of tranquility. This emotional trajectory indicates that the simulation possesses sufficient fidelity to generate engagement and the psychological safety necessary for error to be perceived not as failure, but as data for reflection.
- **Feedback as Scaffolding, Not as Judgment:** By acting under the principle of observational evidence, the automated evaluator positioned itself as a non-punitive "virtual guide teacher". This facilitated an improvement cycle where the transition from "Plan v1" to "Plan v2" was an act of professional self-regulation. The AI's ability to identify subtle weaknesses in mediation allows the pre-service teacher to consolidate technical competencies and build their professional identity in a controlled environment.

7.3 Interface Design as a Cognitive Tool: Text vs. Voice

From an HCI perspective, the results reveal a functional dichotomy that challenges the idea that "more realism is always better".

- **The Value of Text as a Thought Support:** Contrary to the demand for total immersion, the written chat format proved to be a fundamental cognitive scaffolding. By eliminating the pressure of immediate oral response, text forced participants to reflect, organize their ideas, and avoid improvisation. This technical "restriction" favored more conscious and structured decision-making, validating the simplicity of the interface (SUS=80.0) as a facilitator of pedagogical concentration.
- **Voice as Spontaneous Adjustment Training:** On the other hand, the demand for voice interaction should not be understood merely as an aesthetic desire, but as a need to train real-time pedagogical adjustment skills. Voice would allow teachers to practice responding to emotional nuances and subtle mood changes, which is essential for special education teachers who must constantly coordinate with interdisciplinary teams and families.

7.4 The Special Education Teacher as a Conductor of Development

The positive validation from users confirms that this ecosystem aligns with the UNESCO vision [22] and teacher training standards, where AI acts as a support and not a replacement.

- **Professional and Human Identity:** The system does not only teach how to apply strategies for SEN; it places the pre-service teacher in the role of a conductor of a person's development. By interacting with agents that exhibit frustration and progress, the future special education teacher practices the facilitation of social and educational inclusion. This experience strengthens their ability to decide and self-regulate based on specific support needs, preparing them for the complexity of real professional practice where human mediation is irreplaceable.

8 Study Limitations

Despite the positive results in usability and perceived self-efficacy, this pilot study has inherent limitations that must be considered when interpreting the findings and that outline the path for future research.

8.1 Sample Scope and Generalization

The main quantitative limitation lies in the size and composition of the sample:

- **Sample Size:** Although N=17 participants is a statistically robust number for a formative usability evaluation using the SUS scale, it is insufficient to establish definitive generalizations regarding the long-term pedagogical impact on initial teacher education.
- **Gender and Institutional Homogeneity:** The sample was composed entirely of women belonging to a single university institution. This lack of diversity limits the representativeness of the results compared to the global population of pedagogy students.

8.2 Interaction and Cognition Modality Restrictions

The current architecture presents technological boundaries that impact immersion and the development of specific skills:

- **Exclusivity of the Written Format:** Although the chat functioned as a cognitive scaffolding that favors reflection, its exclusivity eliminates critical nuances of real practice. The absence of voice tone and speech rhythm limits training in spontaneous pedagogical adjustment skills, which are essential in special education for responding to the student's subtle moods.
- **Absence of Multimodal Support:** The inability to visualize graphic supports within the chat restricts the fidelity of scaffolding strategies. Since Specific Learning Difficulties (SLD) and Mild Intellectual Disability (MID) profiles require concrete visual mediation, this lack limits the students' ability to practice fully inclusive universal design interventions.

8.3 Diagnostic Delimitation

The study focused on a reduced range of educational needs:

- **Narrow SEN Spectrum:** Only the profiles of Teo (SLD) and Jojo (MID) were validated. Consequently, findings on credibility cannot be extended to conditions of greater clinical complexity that require radically different strategies.
- **Need for Academic Calibration:** A technical limitation was detected in the calibration of the academic competencies of agent Teo, who showed skills inconsistent with his diagnosis of reading difficulties. This discrepancy highlights the need to refine user models to ensure that the agent's academic performance is strictly coherent with their psychopedagogical profile.

8.4 Procedural and Temporal Limitations

The experimental protocol design imposed restrictions on the natural flow of teaching:

- **Bounded Interaction:** The 15-minute time limit, although qualified as an interesting challenge that favors focus, may be artificial. This restriction hinders

the development of deeper closing processes or the management of unforeseen events that, in a real classroom, would demand temporal flexibility that the system does not yet allow to explore.

- **Novelty Effect:** Satisfaction results could be influenced by technological novelty. It is necessary to evaluate whether the observed emotional transition (from anxiety to tranquility) is maintained or evolves toward automated competence in longitudinal use.

9 Conclusions and Future Work

This study demonstrates that a simulation ecosystem based on Generative AI, structured under multidimensional psychopedagogical models, constitutes an effective solution for closing the gap between theory and practice in Special Education. The platform has proven to be not only a content engine but an environment that triggers the fundamental didactic process of planning, implementing, evaluating, reflecting, and adjusting autonomously.

In terms of usability, the SUS score of 80.0 validates that the platform acts as a scaffolding that reduces technical cognitive load, allowing the focus to remain strictly on the pedagogical challenge. The +0.76 point increase in professional confidence confirms that the system offers a "safe space for freedom to fail", where initial anxiety toward practice is transformed into professional security through an iterative cycle of non-punitive feedback.

From an HCI perspective, the agents' ability to simulate a human identity and vulnerability allowed for the development of authentic pedagogical empathy. However, it was identified that while the emotional dimension was highly credible, agent Teo's academic performance showed inconsistencies with his SLD diagnosis, underscoring the need for deeper technical calibration of the user models. For its part, the written chat format proved to be a strategic ally for metacognition by forcing structured and conscious decision-making.

As future work, the system's evolution toward a multimodal platform and a continuous support model is projected:

- **Hybrid Interaction and Visual Support:** Integrating voice technologies to train responses to auditory nuances and mood changes, along with the ability to deploy visual resources that strengthen pedagogical scaffolding in real-time.
- **Profile Calibration and Refinement:** Performing a fine-tuning of the prompts to ensure absolute coherence between the clinical diagnosis and the agents' academic level. Additionally, expanding the catalog of profiles to Early Childhood Education and situations of greater clinical complexity.
- **Transfer Analysis and Traceability:** Investigating the extent to which the acquired security and adjustment skills practiced in the simulation translate into effective performance in real educational centers, consolidating AI as an essential component in the construction of professional teacher identity.

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